Dr. KARIM EMAD
Rispiratory module



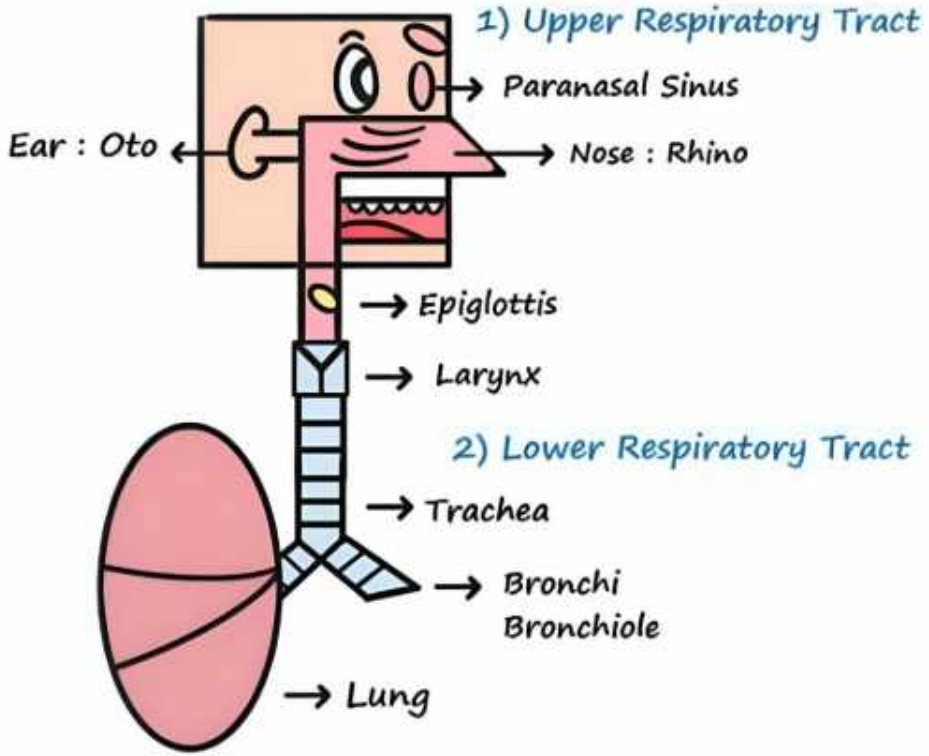
Rispiratory module

Part 1

MICROBIOLOGY

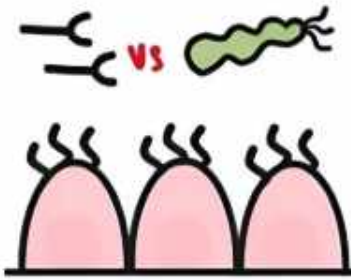
Respirtaory Infection Basics

1- Anatomy + Pathology Basics of Respiratory system

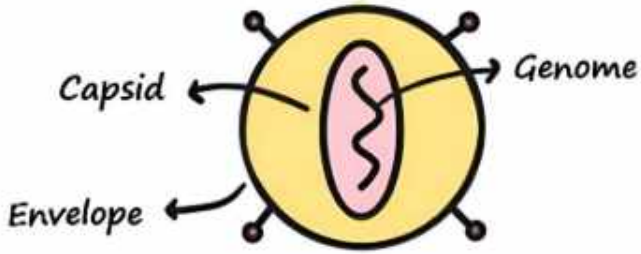


Croup : Laryngiotracheobronchitis

2- NB : Flora of Respiratory Tract :

Upper Respiratory Tarct	Lower Respiratory Tarct
Rich with Flora Cool area with High O2 capacity supprt growth of Microbes Examples of Flora 1- S . aureus 2- S . Pneumoniae 3- Diphtheroid 4- Hemophilus ssp	Sterile " No Microbe in Normal Condition " due to Immune system 

3- Basic of Viral Structure :



4- Respiratiry Tract Infection Scheme : Same as Usual

Dr Karim Mind Map of Infections

Classification of Upper Respiratory Tract Infections

Rhinitis

- virus

1- Rhinovirus

Pharyngitis & Tonsillitis

- Virus

2- Coxsackie Virus

3- Adenovirus

4- Ebstein Barr Virus

5- Cytomegalovirus

- Bacteria

6- Streptococcus pyogenes

7- Corynebacterium Diphetheria

8- Borrilia Venicenti

Epiglotittis

- Bacteria

9- hemophilus Influenza

Sinusitis

- Fungi

10- Mucor , Rhizopus , Absidia

Laryngitis & Tracheitis

- Virus

11- Parainfluenza Virus

Others : Sinusitis , Otitis

1-Common Cold الزكام : البرد

1- Caustive organism : M/C is Rhinovirus " MCQ "

Rhinovirus

Structure : Belong to Picorna Virus

Non enveloped +ve Sense SS RNA virus " Contain Several Serotypes "

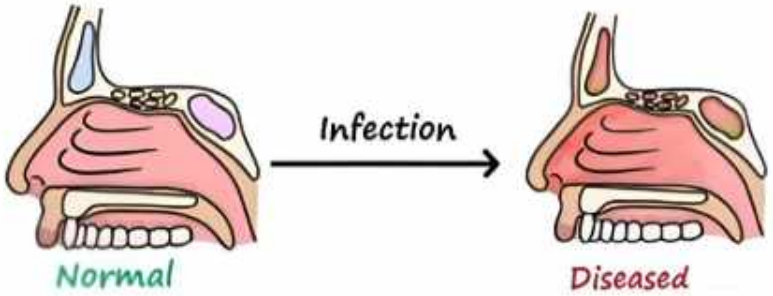
Grow best at 33'C " Infection is limited to URT and Conjunctiva

2- MOT : Droplet Infection and Contact : Direct or InDirect

3- Pathogenesis :

1- Virus attach Nasal Mucosal Cells "
Infection " leads to loss of Cilia and Die

2-WBCs " neutophil , eosinophil , plasma cell , lymphocytes Relaese of Inflammatory mediators stimulate Excess Mucous secrection causing edema and Thickening of Mucosa , Sneezing , Congestion



4- Clinical Picture :



1- Nasal Congestion : Rhinorrhea " Watery Discharge without Purulent "

2- headache , Malaise

3- Decrese sense of Smell

4-Sore throat

Complication :

Local : Otitis media and Sinusitis

Systemic Effect of : Asthma



2nd Infections :

A-Bacterial Pharyngitis or Bacterial Pneumonia

B-Croup خناق

5-Diagnosis : mainly Clinically

6-Prevention : No Vaccine due to Many Serotypes

2-Coxsackie A Virus

1-Structure : Same as Picrona

2-MOT : Same as Common cold

3-Clinical Picture :

Hand , foot and mouth Disease	Herpangina
 Vesicular Leison of Palate , hands and Feet	 Vesicular Papular Rash on Tonsils and Roof of the mouth

3-Adenovirus

1-Structure :

Non Enveloped DS DNA

2-MOT : Swimming Pool

3-Clinical Picture :

Pharyngeoconjunctival	
	
Pharyngitis	Conjunctivitis

Pharyngeoconjunctival Fever " MCQ "

3-Epsein barr virus : EBV

Cause Infectious Mononucleosis



1-Structure : Enveloped DS DNA

2- MOT : Saliva " Kissing "

3-Pathogenesis & C/P : IP : 2 months

Virus Replicate in Nasopharynx then Oropharynx and Salivary gland " Transmit by Saliva "

Then Viremia to the RES " Liver , spleen , BM and B cells "

Cause Infectious Mononucleosis

Clinical Picture of Infectious Mononucleosis

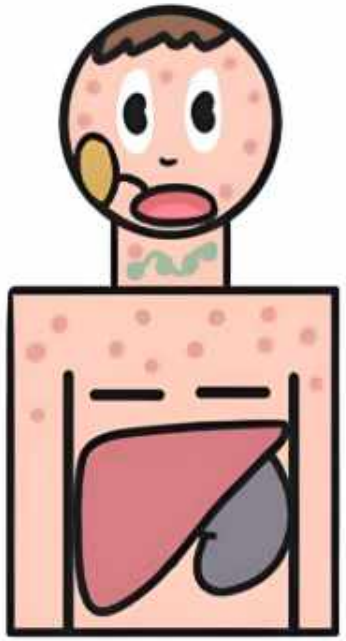
1- FAHM

2- Pharyngitis

3- Lymphadenopathy

4- Maculopapular Rash

5- Heptosplenomegaly with / Without jaundice



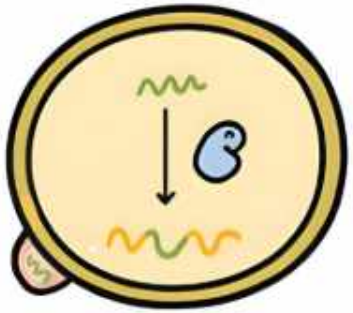
4- Complications

EBV may Infect B cell

CTLs destroy EBV infected B cells and Reducing their Numbers Chatarcteristic " Atypical T Lymphocyte " Found in the Blood



NB : VIP
The virus Remain in the B cells and Oropharyngeal Cell " Integration of the Viral DNA with Host cell Chromosome " result in Tumor Formation



Type	Nasopharyngeal Carcinoma	Burkitt Lymphoma
C/P		

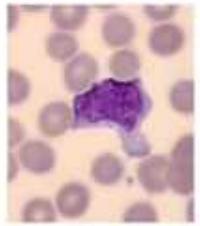
5-Lab Diagnosis

Specimen : Saliva , Blood , Lymphoid tissue

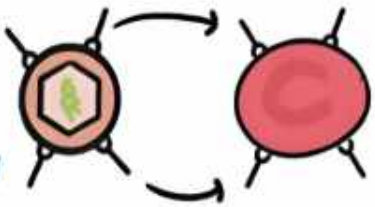
CBC : lymphocytosis with Atypical lymphocyte

PCR : viral DNA

Serology :



Atypical Lymphocyte

Non Specific Ab " Heterophil "	specific Ab
Patient with acute Infection develop transient ab which agglutinate Sheep RBCs Detected by : paul bunnel or Monospot test is +ve 	A-Detect " Viral Capsid Antigen " VCA IgM in Recent Infection B- Detect VCA IgG which persist for life

5-Cytomegalovirus : CMV

Cause Infectious Mononucleosis Like Syndrome

1- Structure : Same as EBV

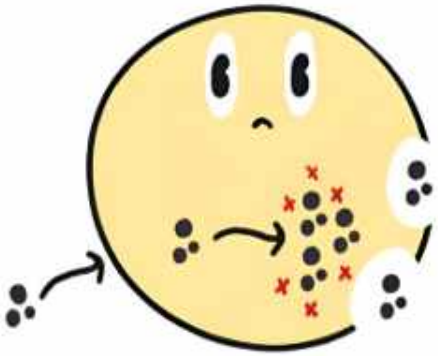
2- Pathogenesis :

Latent in Mononuclear cells to be Reactivated with immune Suppression

3- Complication :

Life threatening pneumonia

4- Lab Diagnosis : -ve Monospot test



6-Streptococcus pyogenes

1-Structure : Gram +ve Cocci arranged in Chains



Virulence factors :



1- Structural :

A-M protein :	B-Hyluronic Capsule :
Major Virulence factor & Antiphagocytic	Antiphagocytic & Non Immunogenic

2- Enzymes : Spreading Enzymes

A-Streptokinase " Fibrinolysin " Convert Plasminogen to Plasmin cause lysis of Fibrin Clots

It's Injected IV to lyse Coronary or Pulmonary Embolism or Thrombus



B- Streptodornae " DNase " : liquifaction of DNA

C- Hyaluronidase : lysis of Hyaluronic acid " Cement Substances "

3-Toxin :

A- Leucocidin

B-Hemolysin " Streptolysin "

Type	Streptolysin S	Streptolysin O
O2 Stability	Oxygen Stable	Oxygen labile
Antigenicity	Non Antigenic	Antigenic
Function	Resposible for B hemolytic activity	Increase B hemolytic Diameter in anaerobic media

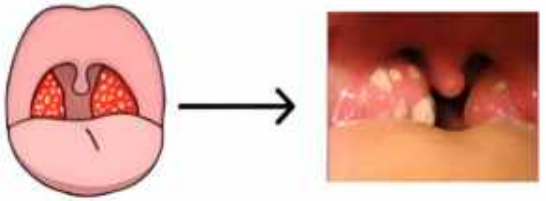
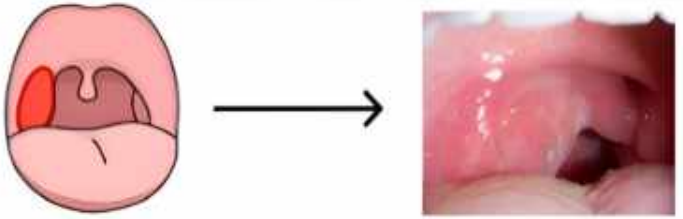
C-Pyrogenic Exotoxin

Type	Type A : Super Ag	Type B
Disease	Toxic Shock Syndrome	Necrotizing Fasciitis

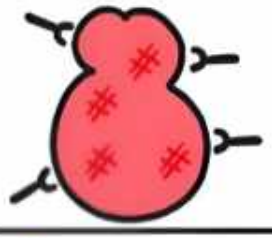
D-Erythrogenic Exotoxin " Super Ag " : Disesae : Scarlet fever

Clinical picture	 Diffuse Erythema	 Strawberry Tongue	 Pharyngitis

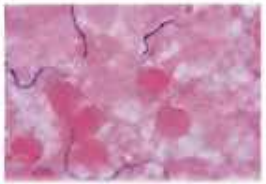
Stages of Streptococcal Pharyngitis :

Type	Acute Tonsillitis	Acute Follicular Tosillitis	Quinsy
			
Clinical picture	Fever , pain , Dysphagia tachycardia , leukocytois Enlarged Cerviacl Ln	Enlarged , Red , Yellow spots " Crypt filled with Debris and pus "	Peritonsillar abscess with deviation of Uvula to Opposite site

Post Streptococcal Complications : Immunological " Hypersensitivity "

A-Rhematic fever	B-Post streptococcal Glomerulonephritis
Type II Hypersensitivity	Type III Hypersensitivity
Auto Ab cross react between Myosin protein of the heart and M protein of S.pyogenes 	Ag-Ab Complex Deposition on the Kidney 

Lab Diagnosis : throat Swab



1- G+ve Cocci in arranged in Chains in Pus



2-Catalase -ve



3-B hemolysis



4- Bacitracin snesitive

5-" ELISA Latex agglutination "

Dr. KARIM EMAD
Rispiratory module



Rispiratory module

Part 2

MICROBIOLOGY

7-Corynebacterium diphtheriae — Cause Diphetheria : الخناق

Acute Communicable disases that Occurs in Children and Become less Common due to Immunization



1-Structure : Gram +ve Bacilli " Non Spore Forming "

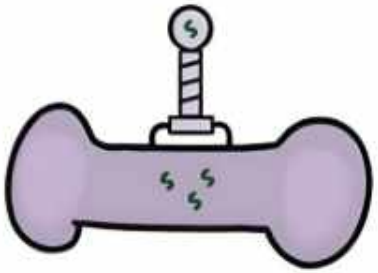
" Coryne : Club shape " شبة المضرب

2-MOT : Respiratory Droplet from Carrier & Case

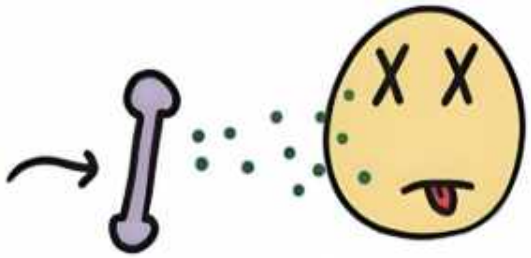
3-Pathogenesis : Virulence Factors : Diphetheria Exotoxin

ability of C.diphtheria to produce Toxin is due to prescense of Prophage " gene of Virus " for Toxin production is Carried on this Prophage the Acquisition of Phage leads to Toxigenicity : Lysogenic Conversion

Inhibit Elongation Factor 2 : EF2 Resulting in Inhibition of the Protein synthesis and Tisse Necrosis



Bacteriophage Insert Toxin gene Into the Bacteria



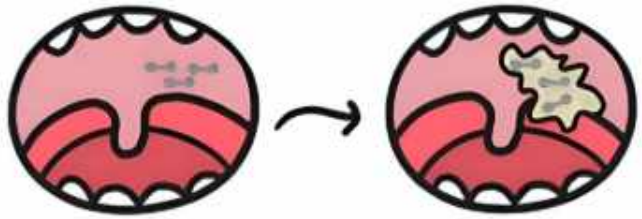
Bacteria Now is Able to Secret Toxin and Kills Cells

4- C/P : IP : 2-5 days

- C.diphtheria Implants on the Mucosa in URT

FAHM + Sore throat " Pharyngitis "

By the Effect of The Toxin causes Necrosis of the Mucosal epithelium leading to Formation of Dirty grey Pseudomembrane with abundant Fibrinolytic Exudate





The Organsim doesn't Penetrate Deeply into mucosal Tissue " NOOOOOOO bacteremia " just Toxemia

5- Compications :

1-Laryngeal Obstruction due to Edema of the Larynx and Perioheral Neuritis

2-Suffocation due to Sloughing and Aspiartion of This Pseudomembrane Leading of Obstruction of airways

3-Absorption of Exotoxins Producing Myocarditis , peripheral Neuropathy



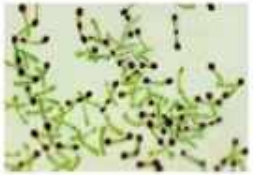
6- Lab Diasgnosis :



Based on Clinical Manifestation + History of Contact lack Immunization + travelling to endemic area

to Differntaied between C.diphtheria and Other Types

Specimen : swab from Throat , nose , any other Leisons



1- Direct Smear : G + Club shaped : bacilli " Chinese letter "



2- Cultivation : Loffler's serum medium



3- K Tellurite media

4- Toxigenicity test : " in refernce lab " Elek's test or ELISA

5- PCR : TOX Gene on Clinical Specimen

6- TTT :

Antitoxin " Life saving " should never be delayed once diagnose Diphethria

7-Prevention :

A- Vaccine Called " DPT " at age of 2,4,6 month route is IM Diphtheria Toxoid + Pertussis Vaccine + Tetanus Toxoid



B- Booster dose Contain Diphtheria & Tetanus Toxoid " DT " only are given at 1.5 years at School age and every 10 years

8-Borrelia Vincenti

Cause Vincent's angina



Commensal Flora in the Mouth

1- Structure :

G-ve Helical , irregular , Loosely , Coiled Bacteria " Spirochetes " + anerobic

2- Pathogenesis & C/P :

in Case of Injury of Mucous Membrane , Immunodeficiency , Malnutrition , Herpes Infection

Borrilia Vincentii with anaerobic Fusiform Bacilli " Fusobacteria " Flourish and Increase in Number

Vincent's angina



Ulcers on the Tonsils and the Gums with Pseudomembrane containing Pus , Necrotic Tissue that Result

Ulcertaive Ginigivostomatitis : Trench mouth

3-Lab Diasgnosis : Swab from Pseudomembrane

Gram stain & Giemsa : Borrilia Vencenti and Fusobacteria & large number of Pus cells

Culture : deep column of Broth enriched with ascitic fluid and Incubated anaerobically

9- Haemophilus Influenzae

Cause Life Threatening Condition " Epiglottis "

1- Structure : Small G-ve rod : Coccobacilli

the Most of Virulent Strains of H.influenza have a polysacchride Capsule the Others are non Capsulated

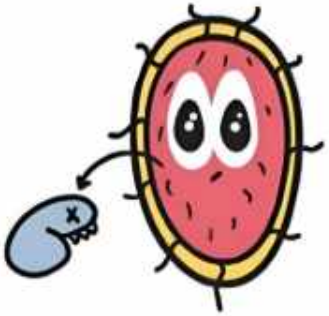
Capsulated Strains are 6 types " a-f " . type b is the Most virulent

Virulence factors :

A- Polysaccharide Capsule formed of polyribitol phosphate " PRP component " Most Important "

B- Ordinary Pili for adhesion

C- IgA protease



Growth requirement it doesnt grow in ordinary media

it needs Growth factor like Hemin and NAD . it Grow in Chocholate agar which contain these Growth factors at 37 c and 5-10 % CO2 " Capnophilic "

H. Influenza Grows well around S aureus agar due to S aureus form Complete hemolysis and Release GF for H.influenza " Satalism "



Satalism

2- MOT : 1- Close Contact with Infected Individual

2- Respiratory Droplet

3- Pathogenesis :



Swell , Block the flow of air into Lungs



Normal

Airway obstruction Severe & High Mortality

4- C/P :

Common Symptoms		
1- FAHM	2- Pharyngitis	3- Irritability
Specific to Children		Specific to Adult
1- Dyspnea 2- Muffled " HoarseVoice " 3- High pitched abnormal Breathing Sound : Stridor		1- Dysphagia 2- Drooling
Potential Lifethreatening Condition :		

5- lab diagnosis : 1- Made Clinically 2- Isolating Bacteria from Blood

6-Prevention : Immunization against H.influnza type B and S.pneumoniae

10-Mucor , Rhizopus & Absidia Cause Fungal Sinusitis

present abundantly in soil and any infection caused by these fungi is named as mucomycosis "العفن الاسود"

1- Structure :

Broad non sepatate hyphae " Filamentous " and Broad Branching



2- MOT : Inhalation of Spores

Risk factors : Immunodeficiency

1- Diabetic Ketoacidosis
2- Leukemia " Lymphoma "
3- Overdose Steroid
4- Severe Burn
5- High serum Iron " Hemachromatosis "

3- Pathogenesis & C/P :

Germination in a nasal passage and Invasion of Hyphae into Blood Vessel causing Thrombosis



Ischemc Necrosis Cause Mass Destruction

The Affected Skin is normal in Early Stage and Become Red , Edematous then Black due to Tissue Death nad Bloody nasal Exudate



4- Complication :

the Disease may progress with invasion of Sinius , Obrit Causing Orbital Cellulitis , eye , Crainial Bone and Brain



5- Lab diagnosis :

A- CT Coronal View : Mucomycosis

B- Specimen from Nasal Discharge & Bisopsy

C- Direct microscopic Examination or Culture & Sensitivity

6- TTT :

A- Correct predisposing factor

B- Surgical Debridement

C- Systemic antifungal : Amphotericin B

11- Human-Para-influenza-Virus : HPIV

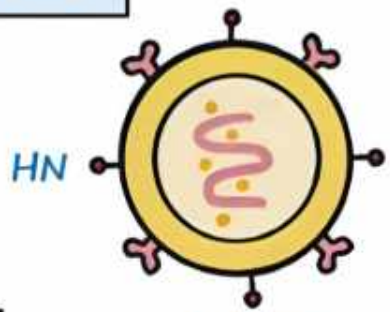
1- Structure : belong to paramyxovirus

A- Genome : -ve sense ss RNA

B- Caspid : nucleocapsid

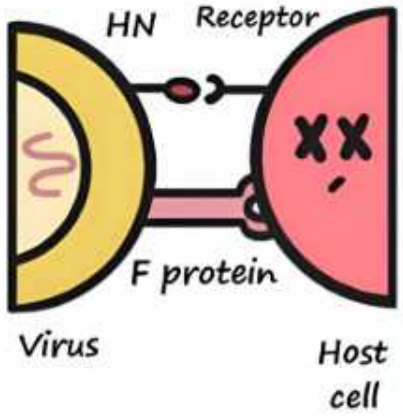
C- Enveloped

Envelope : Lipid enveloped is associated with 2 viral Specific glycoprotein



Classified into 4 Serotypes : HPIV 1 , HPIV 2 , HPIV 3 , HPIV 4

A-Hemagglutinin Neuraminidase : HN protein	B- Fusion : F protein
1- promote the Fusion between the Host Cells leading to Syncytium Fotmation 2- Viral attachmed to RBCs to make Hemadsorption & hemagglutination	promote the Fusion between Host and Viral Cell membrane



2- Pathogenesis :



The Virus attach to Host Cells and Envelope fuse with Host Cell Membrane

The Virus Replicate throughout the LaryngeoTracheobronchial Tree Including Mucous Production

the Vocal cord of Larynx Become Grossly Swollen causing Obstruction of Inflow of air : Inspiratory Stridor

3- C/P : According to age



A- Adult

Limited to URT " Common Cold "

B- Infant and Young Children

Bronchi & Bronchioles & Lung envolved due to : Small Size & relative Immaturity

Diseases Caused by HPIV

HPIV 1 , 2	Croup " Laryngeotracheobronchitis "
HPIV 3	mild to Sever Respiratory illness
HPIV 4	Bronchiolitis , Bronchitis , Pneumonia

4- Lab Diagnosis : Mainly Clinically

Specimen by :



Nasal wash



Broncheoalveolar Lavage



Nasopharyngeal aspirate

1- Direct virus Demonstration : DIF for viral ag

2- Viral Isolation and Identification :
CPE for Syncytia Formation

3-Viral growth is Detected by Hemadsorption by DIF

4- Serotypes is Identified with Specific Antiserum

Seriology : Nt , HI , ELISA test for
Detection IgM or IgG : rising titer

Acute Otitis media Acute infection of Middle ear



M/C oraganism : Strepto Pneumoniae , Moraxella Catalralis , Hemophilus Influenza الثلاثي المرح

Lab diagnosis :

1- Mainly clinical
2- C/S
3- Needle Aspiration of middle ear fluid

Acute Sinsuitis it's acute Inflammation of Paranasal Sinus

M/C oragnaism : Viral + Bacterial الثلاثي المرح

1- Pathogenesis :



if Symptoms worsen after 3-5 days or Persist for longer than 10 years and More severe than normal Experienced with Viral infection a 2nd bacterial infection is Diagnosed

if acute Sinusitis doesnt Resolve , Chronic Sinsusitis can develop from Mucous , Retention , hypoxia blockage of Ostia

2- C/P :

1- Purulent Discharge
2- Nasal Congestion
3- Facial or Sinus Pain
4- Headache

3- Lab diagnosis : Mainly Clinical and by Radiography